

The Rail Pathfinders

Cheah Kika | Jin Donghua

Zeph Tang Zhenkai | Zhang Yizhuo



Commute Copilot

A little help along the way.

Persona: Mdm Lim, the occasional traveller

Our primary design target is Mdm Lim from the [competition persona brief](#): an occasional traveller who walks slowly, avoids stairs and needs to plan an unfamiliar journey in advance. This is a design choice, not a finding from a user study.

We prioritise adjustable large text, Simple guidance, clear current and next steps, advance route review and a Show to staff message. Walking preferences and a read-only trip link support planning and assistance. The traveller stays in control of route changes. Our tested Paya Lebar-to-Bugis journey demonstrates these interactions; it does not validate Mdm Lim's hospital journey or prove that its paths are accessible.

Architecture: how a route becomes guidance

Browser: HTML/CSS and JavaScript collect endpoints, time and preferences. A cancellable Web Worker runs the rail/bus routing engine over packaged schedules, bus patterns and walking links. Matching demo exclusions are applied before route ranking. Leaflet displays the route using attributed OpenStreetMap-derived geometry.

State and services: an accepted-journey state machine separates suggestions from active guidance. LocalStorage retains preferences, saved routes and progress until removed; a service worker caches the app and prepared guidance. A Cloudflare-compatible Worker serves the app and sharing API. Read-only trip links use persistent D1 storage. Optional server adapters call OneMap and LTA with credentials kept off the client. Deterministic rules perform the routing; no LLM is required.

Assumptions we make explicit

The user supplies the correct endpoints, journey date and any deadline, and confirms actual progress. We interpret travel times in Singapore time and use service calendars within the packaged data's coverage. Walking allowances and preference presets are editable design defaults, not measured personal walking speeds.

The demonstration treats saved closures/cancellations as valid only for their stated service, direction, segment and time window. Simulated incidents are labelled. Missing live data means unknown or unavailable; it is not evidence of normal service. Offline guidance requires a successful readiness check while online.

Limits and checkable evidence

Known limitations

Routing and data: schedules are dated; bus ride/wait times and fares are estimates. Demo closures can change routes, but official notices remain advisory and delay scenarios do not produce validated revised arrivals. We have not established prediction accuracy, real-world time savings or superiority to another journey planner.

Incomplete integrations and accessibility: the submitted hosted configuration lacks OneMap and LTA arrival credentials, so address/GPS routing and live arrivals are unavailable. The demonstration uses station or bus-stop endpoints. Continuous step-free indoor paths, endpoint walks and current lift availability are unverified; advance lift warnings are not demonstrated. The accessibility preset blocks starting an unverified route. These remain gaps against Mdm Lim's needs; the demonstration does not certify an accessible journey. Crowding, shelter and cycling are also unverified.

Use and privacy: desktop automation and mobile emulation do not replace physical-phone or commuter trials. Coordinates cannot confirm a floor, boarding or arrival. Offline maps/live feeds are not guaranteed; journey notifications are off. Sharing sends planned endpoints and route geometry, not ongoing device-location tracking; access can be cancelled or expires, while inactive-server cleanup is not guaranteed at an exact time.

What a judge can reproduce

Automated checks: 555 JavaScript and 44 Python test cases. These are the runner totals from the recorded npm run check execution, not percentages or counts of users/journeys. The Python total is **27 rail-import + 16 bus-import + 1 walking-metadata**. All passed; the same command also built the app. The [recorded results](#) and [exact test commands](#) are pinned to the source snapshot.

To check those totals: download the [pinned source](#), follow its README prerequisites, then run `npm ci` and `npm run check`. Compare the Node summary and the separate Python summaries; browser checks are additional and are not included in these totals.

Route-change demonstration: use Paya Lebar (EW8) to Bugis (EW12), 21 September 2026, 10:00 SGT. These are chosen test inputs. Record the baseline, save a whole-line EW closure covering that day, and search again. Verify the suggested route avoids EW and can be explicitly started. Reload and repeat the same search to verify that the saved closure still changes the suggestion. Resolve the incident and verify the baseline becomes eligible again. The appended guide gives the clicks; [the browser test](#) checks the route IDs and accepted state. This proves fixture behaviour, not performance against live disruption outcomes.

Try the app: [Commute Copilot](#) | **Source:** [The Rail Pathfinders](#)

The illustrated user guide follows. Application source: f5fb04b9d5bb; reproducible source snapshot: d787c8d6501b (19 September 2026).

A little help along the way.

An illustrated guide to Commute Copilot for NEBULA X PS2 judges.

Commute Copilot

Plan a journey. Understand the next step.
Keep control of progress and sharing.



SCAN TO OPEN ON A PHONE

<https://commute-copilot-nebula.simhongmen.chatgpt.site>

Android: Chrome | iPhone: Safari | No account or installation required

USE THIS DATED EXAMPLE

Tampines (EW2) to Bugis (EW12)

Depart later | Monday 21 September 2026 | 10:00 SGT

THE 3-5 MINUTE JUDGING PATH

0:00-1:30

Plan and accept

Open the link, select the dated route, compare and start. Pages 2-4.

1:30-3:00

Follow the journey

Confirm a step, try Simple guidance and show the staff card. Pages 5-6.

3:00-5:00

Explore useful extras

View station facts, save a commute and inspect sample spending. Pages 7-9.

A PROTOTYPE WITH CLEAR BOUNDARIES

Screenshots show real app screens using retained transport data. The dated example is for judging; it is not a claim of live transport or verified indoor navigation.

Start on your phone

Use a normal browser tab. The QR code and link on page 1 open the same HTTPS app.

1 Open the app

Connect to the internet. Open the link in **Chrome on Android** or **Safari on iPhone**. No APK download, sign-in or mandatory Home Screen install is needed.

2 Choose your preferences

In **Settings**, choose a schedule style, walking limit and route priority. For this example, leave **Require a verified step-free route** unchecked.

3 Select both stations

In **Plan**, type Tampines and Bugis. **Tap each matching suggestion** so the app confirms the station.

4 Set the example time

Tap **Depart later**. Set **21 September 2026, 10:00**. Confirm the calendar with **Use this date**. Times are in Singapore time.

5 Search

Scroll down and tap **Find my route**. Keep the dated example even if judging happens after the data coverage ends.

If older controls remain

Refresh, wait for the update to load, then refresh again. Do not clear browser data unless you intend to remove saved routes and spending.

Selected stations and the fixed judging date. App capture.

PHONE SETUP

Optional: use the browser's Add to Home Screen action. Browser use is sufficient. Physical Android/iPhone testing is not yet recorded; screenshots use a mobile-sized browser viewport.

Choose the route that fits

Route cards make arrival, transfers and walking easy to compare.

1 Read the route card

Check the line or service, total duration, arrival time, transfers and walking. These are planned or estimated values, not a live vehicle position.

2 Choose an option

Tap the route card you want. If several options appear, compare them using your walking and timing preferences.

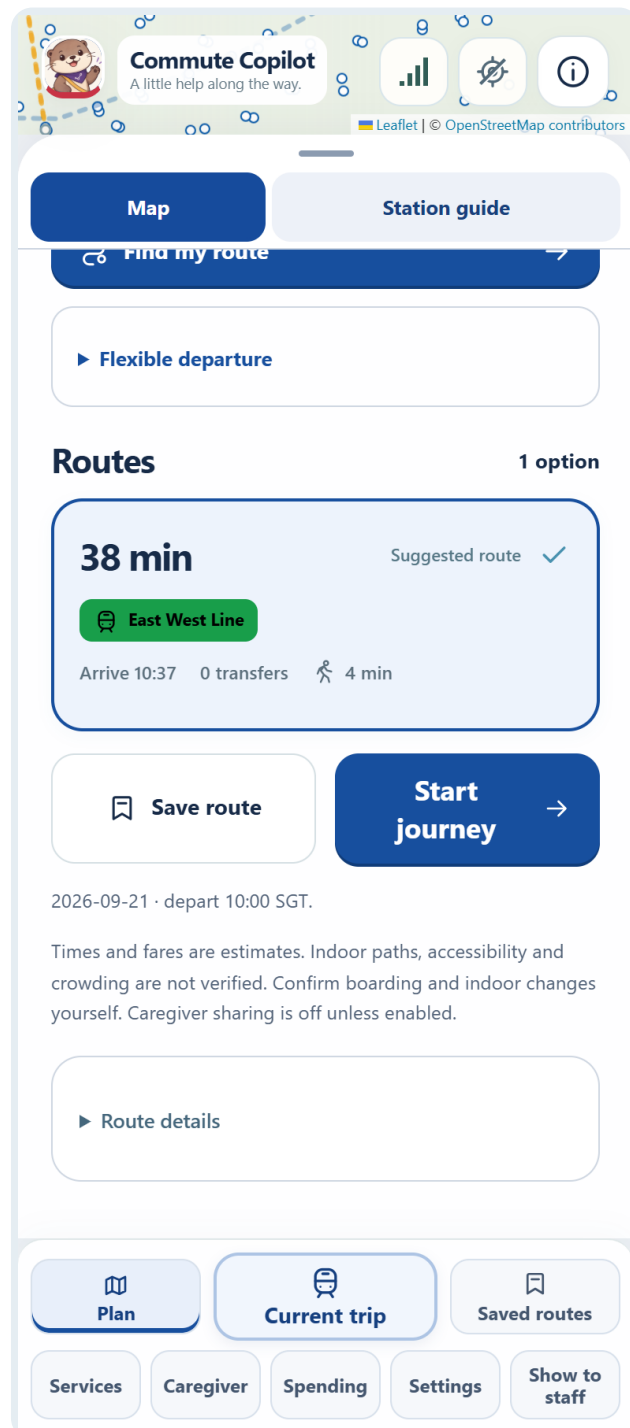
3 Review details

Open **Route details** for source and timing information. **Flexible departure** can compare departure choices.

4 Start when ready

Start journey starts the selected route and opens Current trip. Review **Route details** first. **Save route** stores the endpoints and preferences for later.

For this example, the screenshot shows the East West Line with no transfer. Use what the current screen displays; do not expect every search to have alternatives.



An app result. Start journey begins guidance for the selected route.

USEFUL FOR JUDGING

Searching, selecting a route or saving it does not start a trip. The traveller reviews Route details, then explicitly presses Start journey.

Check the steps, then start

Route details explains the whole journey before you start guidance.

1 Open Route details

In the selected route result, check the start, destination, date and planned arrival. Scroll through the walking, waiting and ride steps.

2 Inspect the timeline

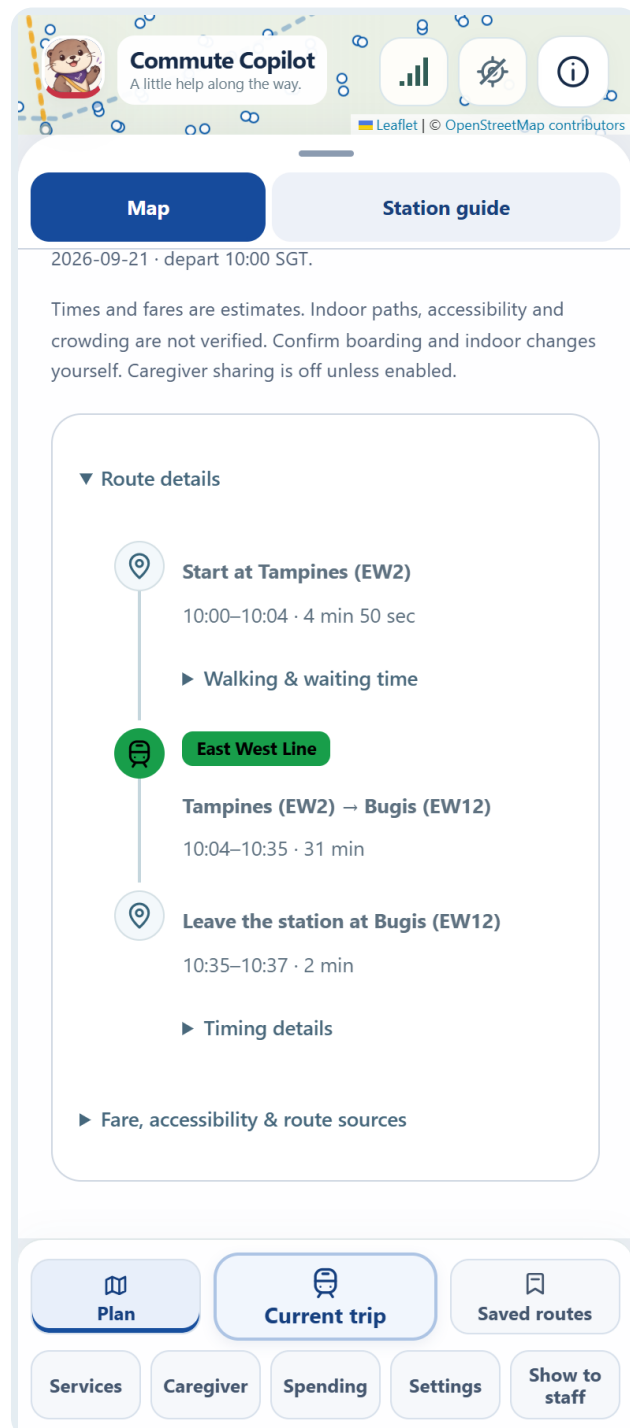
The example starts at Tampines, takes the East West Line and leaves the station at Bugis. Expand timing details when you need more context.

3 Keep location optional

Location assistance is separate from caregiver sharing. You can demonstrate and operate the core trip with manual progress and no location permission.

4 Tap Start journey

This accepts the route and opens **Current trip**. An accepted trip stays available while you use other tabs and after a reload.



The Route details timeline before the journey starts.

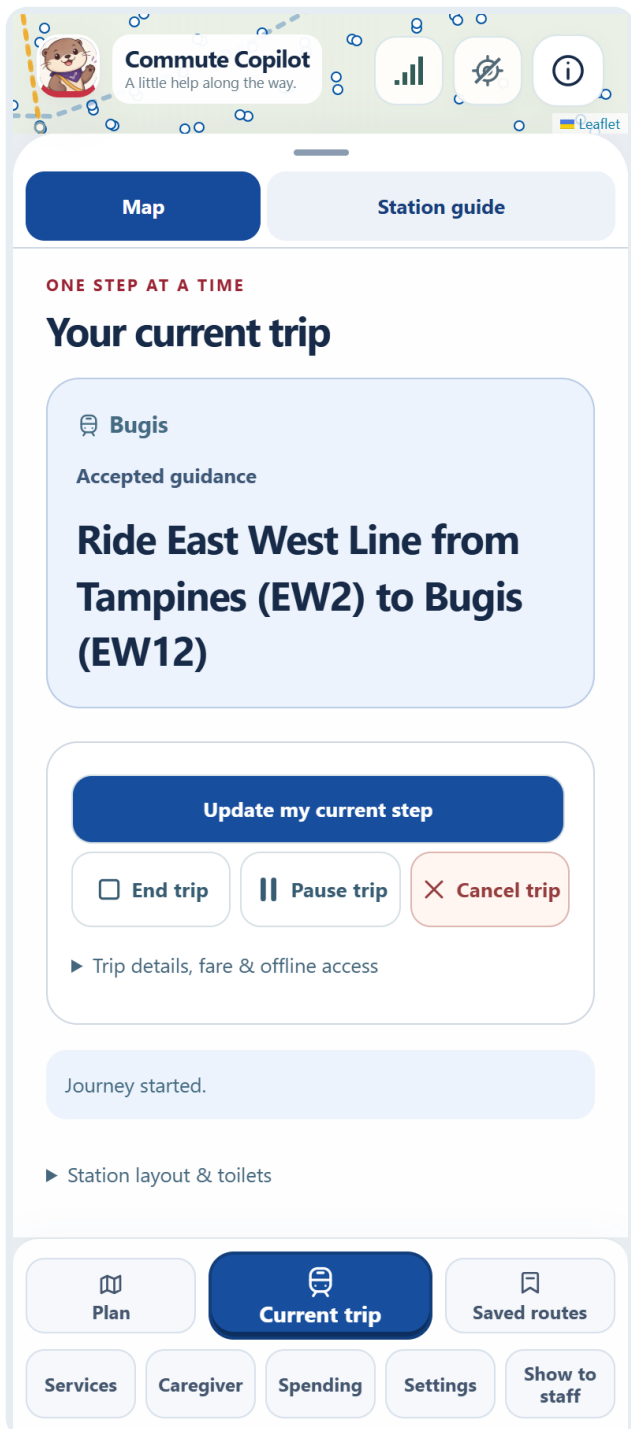
YOU REMAIN IN CONTROL

The app does not infer that you boarded or arrived just because time passed. Confirm progress yourself. GPS is optional assistance, not proof that you reached a checkpoint.

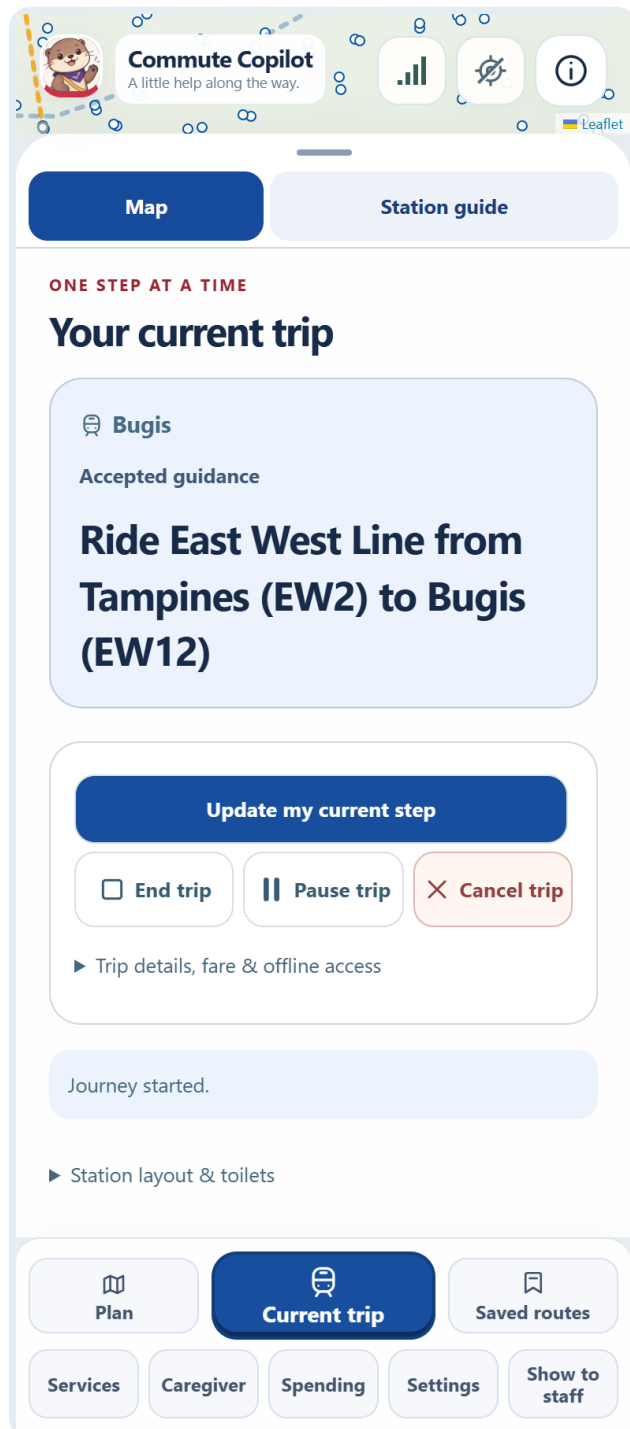
One step at a time

Confirm progress manually, then try a simpler presentation of the same accepted route.

CURRENT TRIP



SIMPLE GUIDANCE

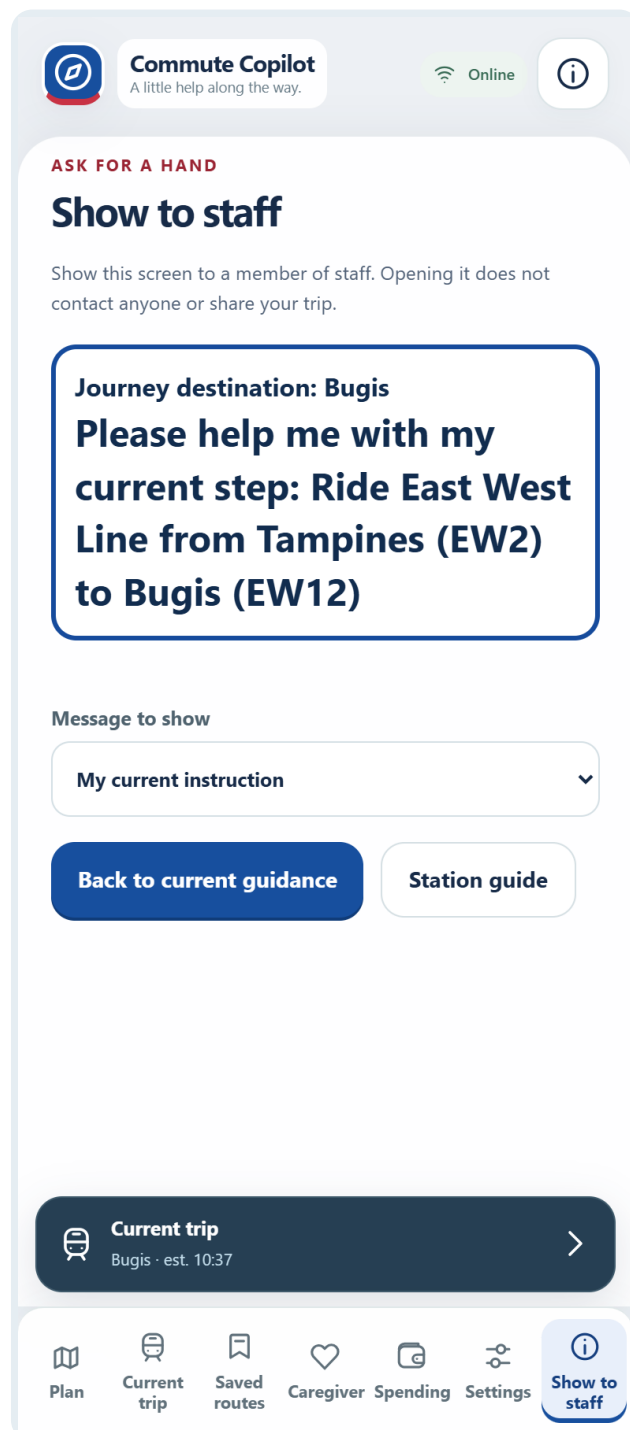


1. Open **Update my current step** (or **Confirm my current step**), select the step reached, then tap **I am here**. 2. In **Settings**, enable **Simple guidance** and return to Current trip. 3. **End trip**, **Pause trip** and **Cancel trip** share one row. End only on arrival or in a labelled demonstration. Location controls are in the top bar.

Show the instruction to staff

A large, relevant message helps explain what you need without retyping the journey.

- 1 Open Show to staff**
Use the bottom navigation or the button in **Current trip**.
- 2 Choose the message**
Use **Message to show** to request help with the current instruction, correct stop or platform, transfer, exit, lift access or a stop alert.
- 3 Show your phone**
Present the large card to a member of staff. The destination and relevant instruction provide context.
- 4 Return to guidance**
Tap **Back to current guidance**, or open **Station guide** for the available operator facts.



The staff card reflects the traveller's accepted journey and current step.

WHAT THIS ACTION DOES

Opening the staff card contacts nobody and does not share the trip. It is a screen to show in person, not an assistance booking or an outgoing message.

Use the station guide

Read available facility information, then confirm your route with station signs or staff.

1 Open Station guide

Tap the **Station guide** tab above the journey panel. You can reach it while a trip is active.

2 Read the location facts

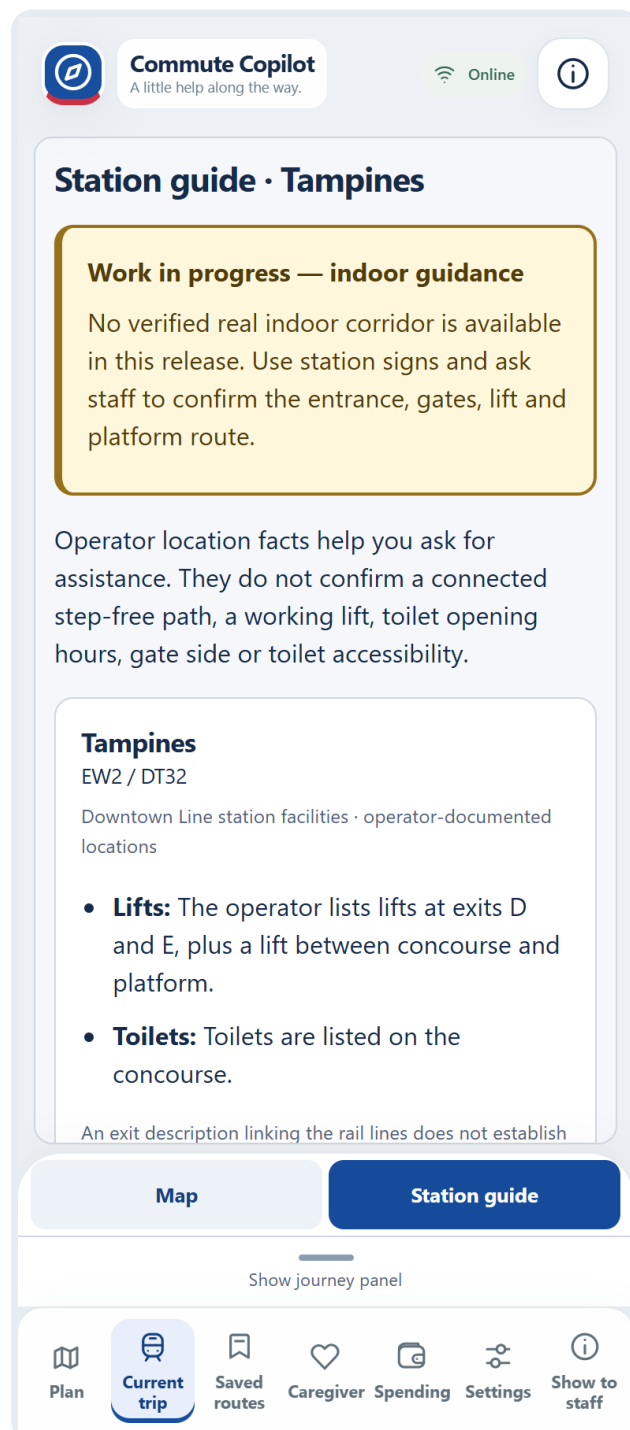
For supported stations, the app presents operator-documented facility locations such as lifts and toilets. Tampines is shown here.

3 Notice the evidence limit

The **Work in progress** notice is part of the product. There is no verified real indoor corridor in this release.

4 Make space when needed

Tap or drag the journey-panel handle to resize it. The **Map** and **Station guide** controls remain available.



The warning and operator facility facts are both visible in the actual app.

ACCESSIBILITY BOUNDARY

Facility facts do not guarantee a working lift, a connected step-free path, an accessible toilet or a verified entrance-to-platform route. Confirm the route with station staff.

Save and pin a commute

Keep familiar endpoints and preferences in the browser you are using.

1 Save after searching

From route results, tap **Save route**, give it a useful name and choose **Save Route on this device**.

2 Open Saved routes

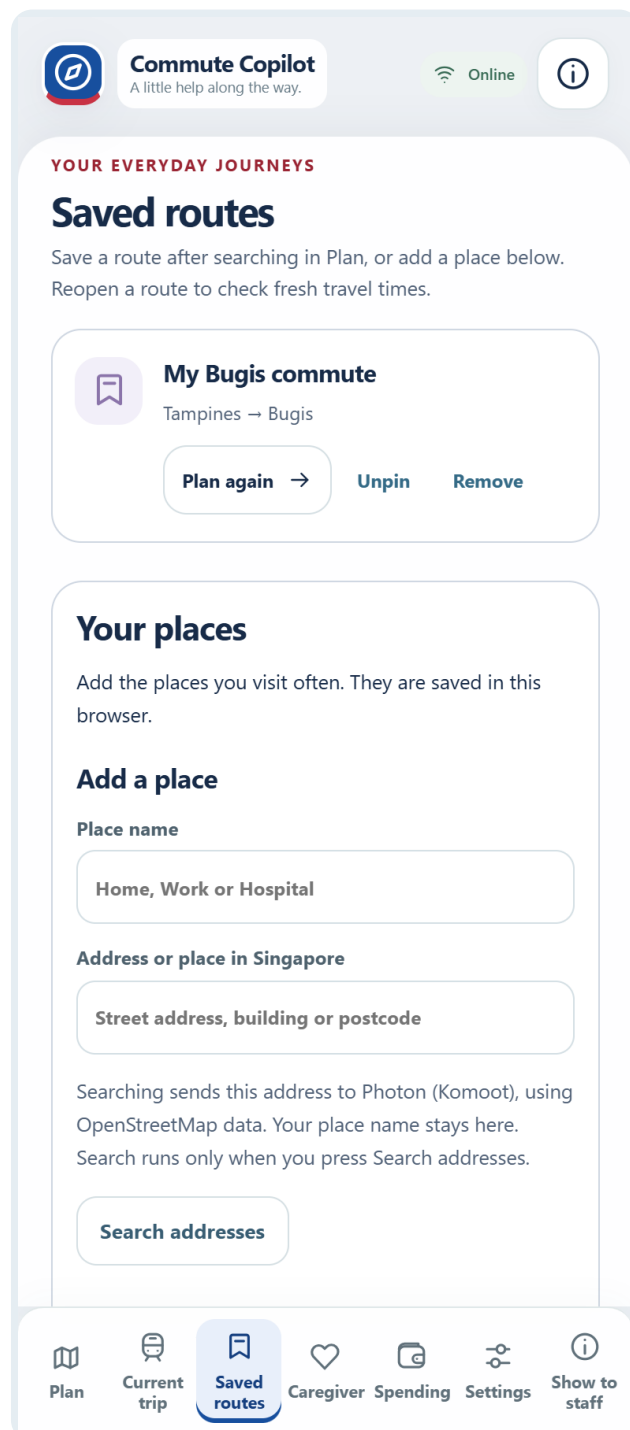
Find the saved trip. **Plan again** returns to planning; **Pin commute** makes it available as a shortcut.

3 Use Your commutes

In **Plan**, up to four pinned or recent shortcuts can appear. **Check routes** and **Reverse** calculate fresh Leave now options.

4 Replay the judging date

For this guide, use **Plan again** and reselect **21 September 2026, 10:00** if the current date is outside coverage.



The saved example is already pinned, so the action reads Unpin.

SAVED ON THIS DEVICE

Saved routes, places and preferences belong to this browser and website address. They are not a cloud account. Replanning never automatically starts or replaces a trip.

Review spending and a budget

Completed journeys can be reviewed without treating estimates as confirmed charges.

1 Finish a journey

Use **End trip** only at completion, or for a clearly labelled demonstration. The completed journey is recorded once.

2 Open Spending

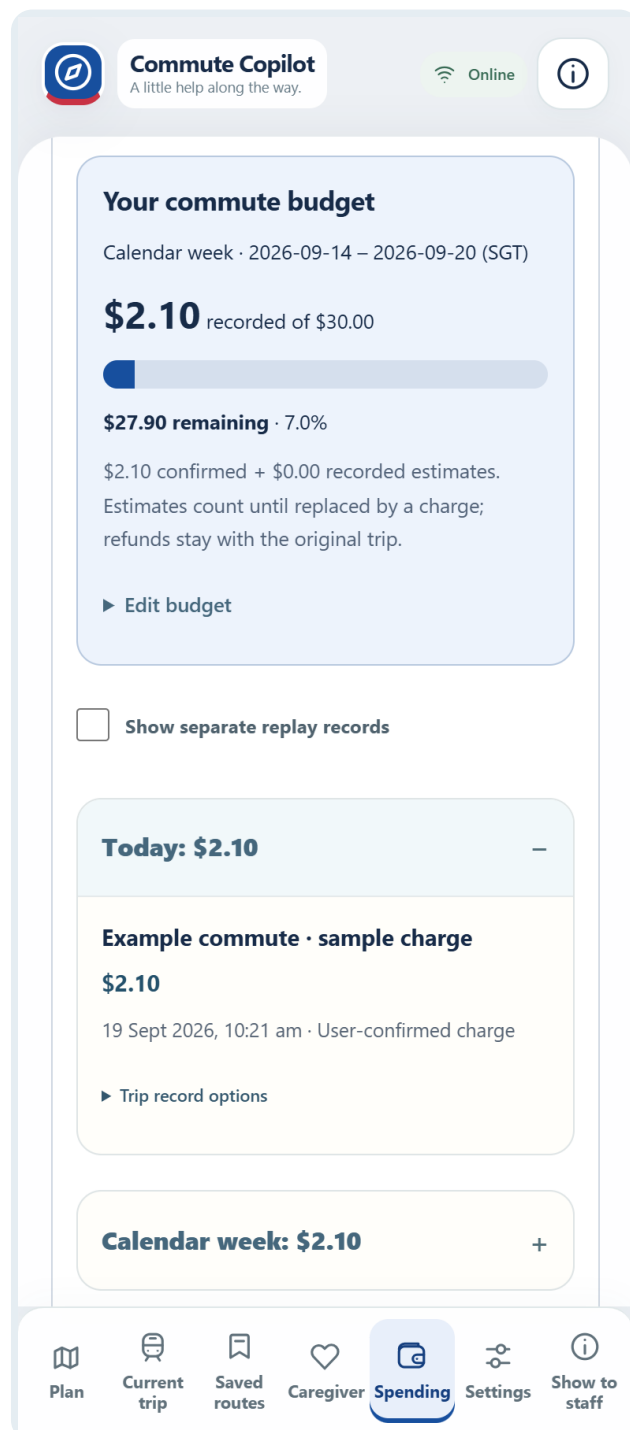
Review the journey record. Enter or edit the amount actually charged when known. A fare estimate can be replaced by the confirmed charge.

3 Set a budget

Tap **Set a budget**, choose week or month, enter an SGD amount and save. Use **Edit budget** to change it later.

4 Interpret the totals

The app separates confirmed charges, estimates and records awaiting an amount. Missing prices make the remaining budget incomplete.



EXAMPLE DATA: a manually entered SGD 2.10 charge and SGD 30 weekly budget.

DEMONSTRATION AMOUNTS

The SGD 2.10 record and SGD 30 budget were manually entered as sample data. They are not a verified fare for Tampines to Bugis. The app does not process payments.

Offline use and sharing

Prepare saved instructions online. Sharing remains an explicit, separate choice.

Prepare offline access

While online, accept a trip. In **Current trip**, open **Trip details, fare & offline access** and tap **Check offline availability**. Continue only after the app confirms readiness.

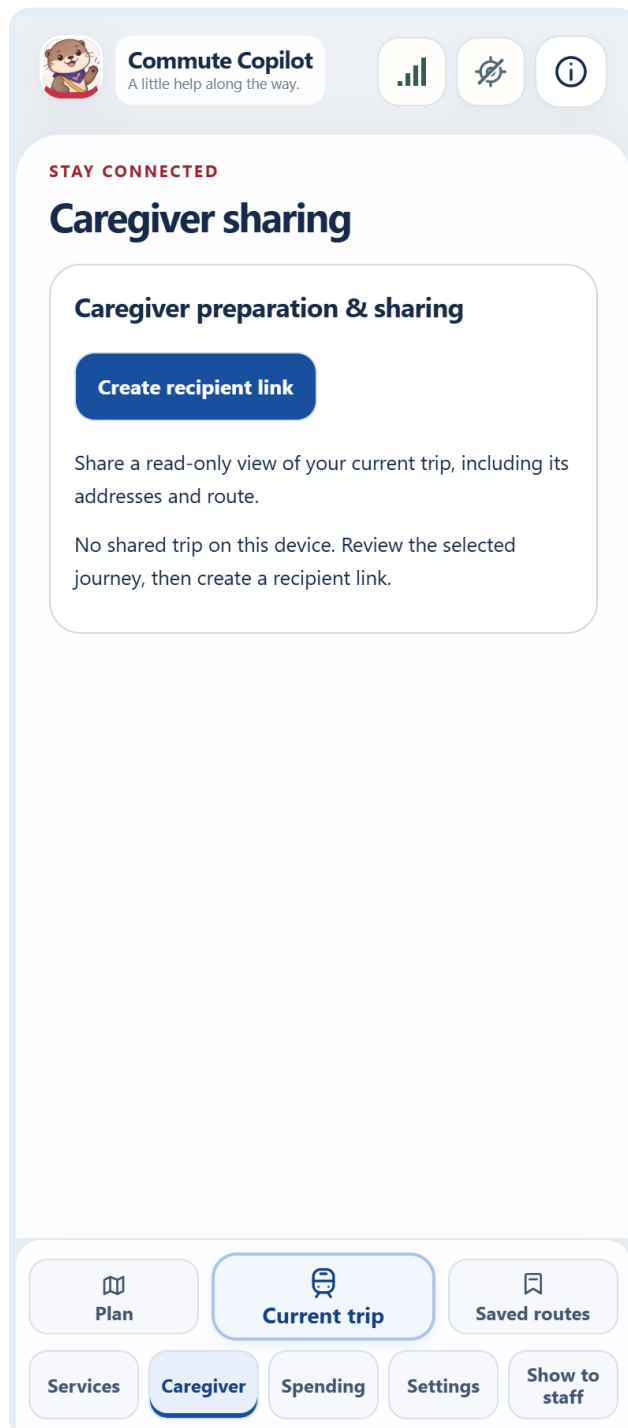
The accepted instructions and saved progress can then restore. Fresh address searches and live services need a connection. Map tiles are not guaranteed offline.

Caregiver permissions

Review your journey, then use **Create recipient link** to share a read-only trip view. Send the link only to your chosen recipient; it includes the trip addresses and route.

As the creator, **Refresh shared trip** publishes your currently selected plan; **Cancel sharing** removes the shared view. The recipient uses **Refresh trip** to fetch the latest published view. Sharing remains optional.

The link includes planned endpoints and route geometry, without ongoing device-location tracking. Do not rely on closed-tab alerts or native Dynamic Island support.



Current caregiver controls create a read-only view of the selected trip.

OFFLINE DEMONSTRATION

Offline readiness requires HTTPS (or localhost) and a working service worker. Use the app's check first. Physical-phone airplane-mode behavior has not been verified.

Know what the prototype covers

Core station and bus-stop planning uses packaged data and works without provider keys.

Read the route map

Walking uses dotted grey paths. Bus routes use blue with white centre dashes. Trains use solid line colours. Badges identify lines or services; dots mark transfers.

Geometry is approximate. Pinch or scroll on the map to zoom it. Use **Settings** to enlarge app text. Use the panel handle to reveal more map; select a bus stop to inspect its services.

Use the dated data window

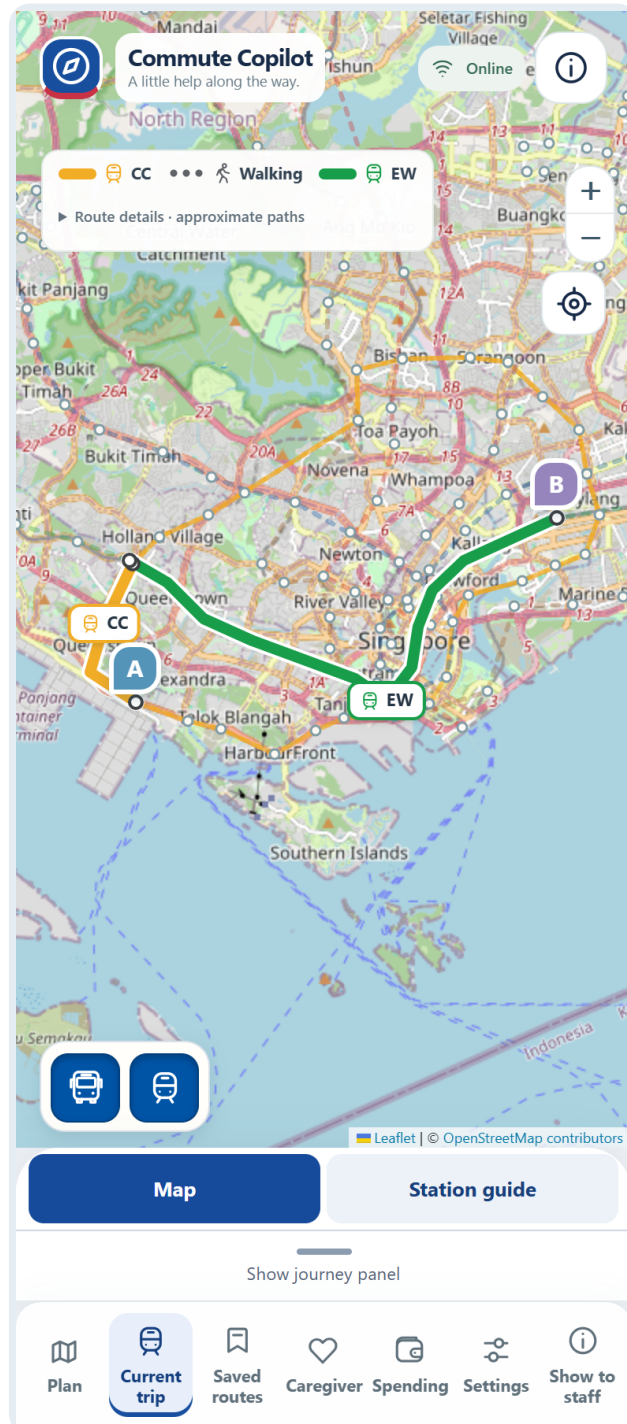
Rail: 18 September to 31 December 2026, subject to service calendars.

Bus: 18 September to 2 October 2026. Bus frequency and riding times are estimates, not vehicle timetables or live arrivals.

For bus travel, select bus-stop suggestions in Plan. Directory presence does not guarantee a route for your selected time. Check **Where can I travel?** or **Demo & coverage** for limitations.

Services needing configuration

Street-address itineraries and live bus arrivals depend on provider availability. For the disruption test, select station or bus-stop suggestions so saved closures can be enforced.



Haw Par Villa to Aljunied via Buona Vista; Fewer transfers, 15 min extra.

TEST A DISRUPTION

Saved unresolved closures affect suggested routes. Pages 13-14 show how to save a fictional closure, return to the app, test the alternative and start guidance. This is a demo, not a live disruption feed.

Find a stop and its services

Search across the included stop directory, or explore nearby stops on the map.

1 Search by name or code

In **Plan**, type a bus-stop name or stop code in an endpoint field. **Select a matching suggestion** to confirm the stop.

2 Open the stop details

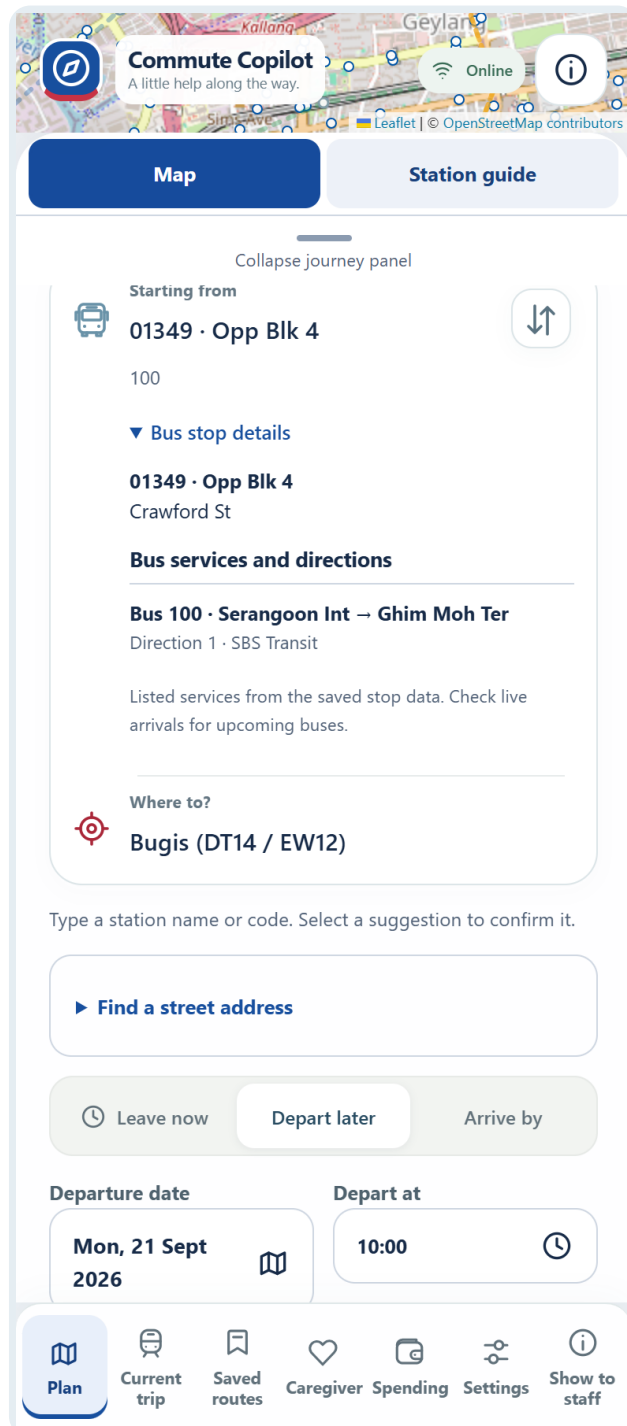
The selected stop shows compact service numbers. Tap **Bus stop details** to inspect directions and operator information.

3 Explore the map

Tap **Bus stops (bus icon)** and zoom to a neighbourhood. **Train network (train icon)** shows or hides the background rail network. Select a stop, then use **Start here** or **Go here** to set an endpoint.

4 Check a timed route

Select the other endpoint and tap **Find my route**. Some directory services have no timed estimate, so a listed service may not produce a route for your date and time.



The selected stop shows its services, travel directions and operators.

DIRECTORY AND ROUTING ARE DIFFERENT

Directory: 5,208 stops. Timed routing: 5,206 stops, 776 directional patterns and 583 service numbers. The map shows up to 450 visible stops at once; the complete directory remains searchable.

Create a demo closure

A repeatable test: Paya Lebar (EW8) to Bugis (EW12), 21 September 2026 at 10:00 SGT.

1. Plan the normal route first

In **Plan**, select **EW8 Paya Lebar** and **EW12 Bugis** from the suggestions. Choose **Depart later**, set the date and time above, then **Find my route**. Note the EW route. Leave it unstarted for this test.

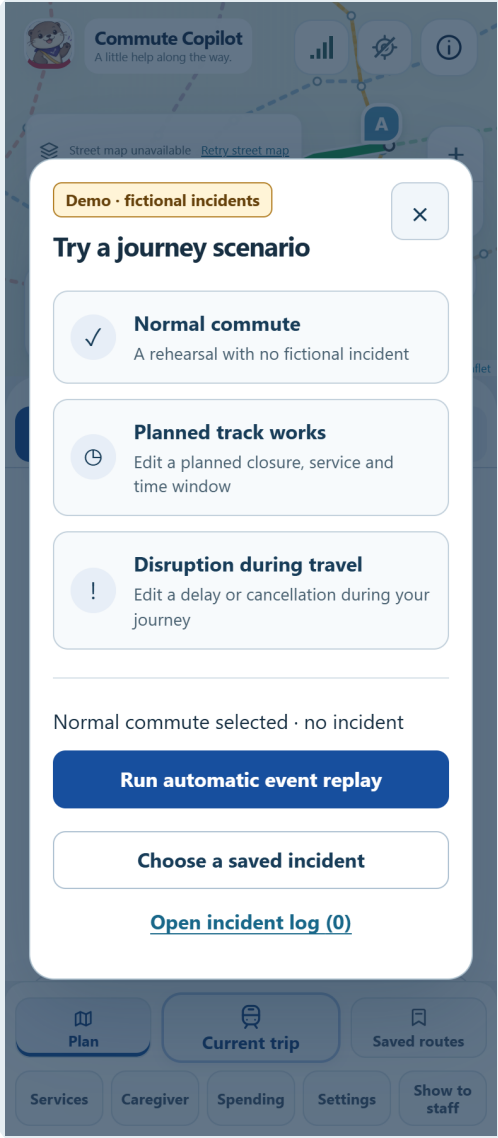
2. Open the demo controls

Tap the top-right information button (**Try a journey scenario**), then **Planned track works**. Enter these fictional details:

Incident title	Demo EW line closure
Incident type / Severity	Closure / Severe
Service or line	EW
Route scope	Whole service or line
Direction	Both directions
Starts at (SGT)	21 Sep 2026, 00:00
Ends at (SGT)	21 Sep 2026, 23:59
Incident details	Fictional closure of the EW line for this demo.

3. Save and confirm

Tap **Save to incident log**. Confirm the demo entry is **Active**. Scroll within the dialog to reach the save button when needed.



The information button opens these demo controls. Choose a saved incident fills the available width.

Open incident log opens the full log page. It stores fictional incidents in this browser on this device.

USE A CLOSURE FOR THIS TEST

Closure and Cancellation exclude matching services. Delay is advisory. Incidents must overlap the chosen journey date and time. Existing unresolved incidents also apply; resolve unrelated entries before this test.

Return, check, then start

Saving a closure changes suggested routes. Starting the alternative is a separate, explicit action.

4. Return to the main page

In the dialog, tap the **X (Close demo dialog)**. On the full incident-log page, tap **Back to Commute Copilot** at the top. Open **Plan**; reselect the same stations, date and time if needed.

5. Confirm the disruption is reflected

Tap **Find my route**. Look for **Demo incidents active**. The suggested route should change and avoid EW trains. Check the line badges, **Route details** and map. A save in the open app also refreshes an existing search.

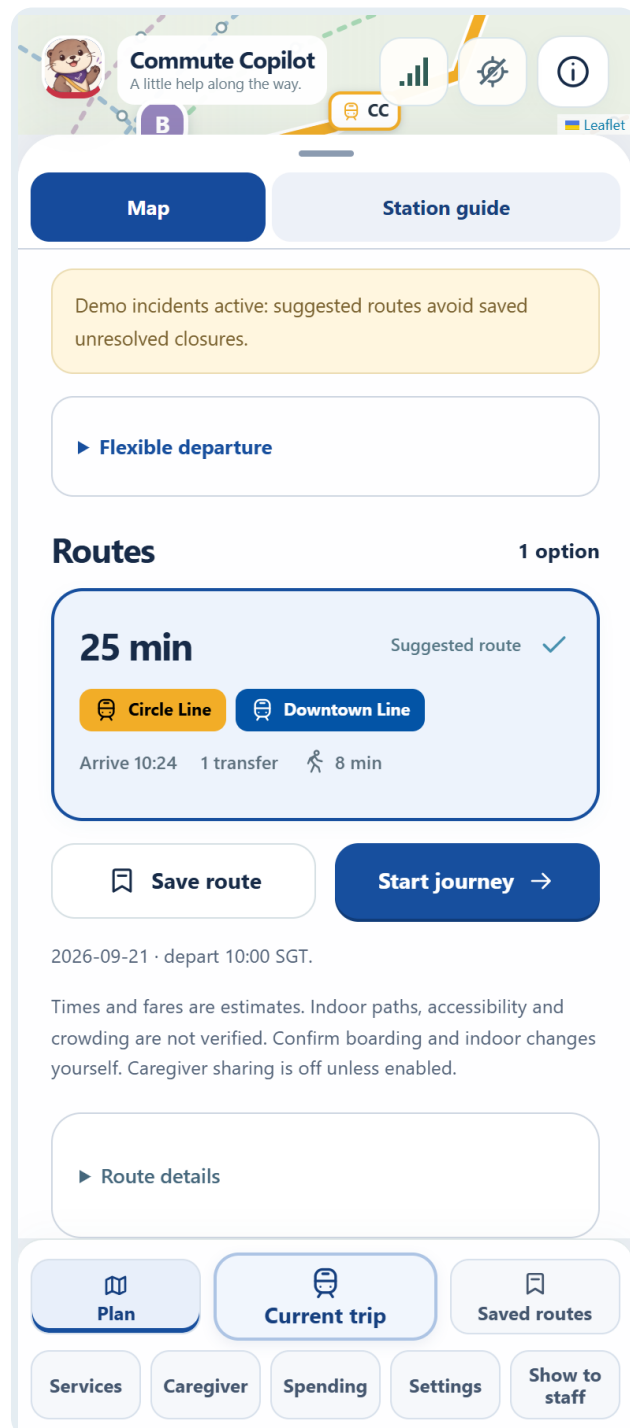
6. Activate the alternative

Select the alternative card and tap **Start journey**. **Current trip** opens with its directions. **Run automatic event replay** only previews a comparison; it does not replace an accepted trip.

7. Resolve and test recovery

Open the information button, **Choose a saved incident**, then **Mark resolved**. Close it and search again. The original route becomes eligible again. **Normal commute** does not clear saved incidents.

Optional reset: **Clear demo log**, then **Delete demo log**, removes all demo entries from this browser.



The suggested alternative avoids EW trains during the saved closure.

IF A TRIP IS ALREADY ACTIVE

Use Current trip > Cancel trip > Yes, cancel trip before starting another. Then Plan > Find my route > Start journey. For a mid-trip rehearsal, set the origin to your confirmed current station or stop.